

HAILAR, China

WMO: 505270

Lat: 49.2500N

Lon: 119.7000E

Elev: 650

StdP: 93.76

Time Zone: 8.00 (E08)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.1	(c) -33.8	(d) -39.5	(e) 0.1	(f) -35.9	(g) -36.8	(h) 0.1	(i) -33.6	(j) 9.8	(k) -19.0	(l) 8.4	(m) -20.2	(n) 2.0	(o) 180	(p) 0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 31.2	(d) 18.1	(e) 29.3	(f) 17.4	(g) 27.5	(h) 17.0	(i) 20.5	(j) 26.9	(k) 19.6	(l) 25.7	(m) 18.7	(n) 24.4	(o) 4.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.5	23.2	17.5	13.6	22.3	16.6	12.8	21.6	62.5	26.9	59.1	25.7	56.0	24.2	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.4	(b) 9.0	(c) 7.7	(e) -37.8	(f) 34.3	(g) 3.1	(h) 2.6	(i) -40.0	(j) 36.2	(k) -41.8	(l) 37.7	(m) -43.5	(n) 39.1	(o) -45.8	(p) 41.0		
(a) 10.4	(b) 9.0	(c) 7.7	(e) -37.9	(f) 22.0	(g) 3.0	(h) 1.5	(i) -40.1	(j) 23.0	(k) -41.9	(l) 23.9	(m) -43.6	(n) 24.7	(o) -45.8	(p) 25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-25.1	-19.9	-9.8	3.1	11.4	18.2	20.9	18.7	11.7	1.7	-11.6	-21.9	(6)
(7)		DBStd	16.93	5.16	5.58	6.63	5.09	4.55	3.93	2.81	3.24	4.53	5.40	6.77	5.41	(7)
(8)		HDD10.0	4722	1089	837	614	214	37	1	0	0	34	261	648	987	(8)
(9)		HDD18.3	6923	1348	1071	873	457	219	50	8	33	205	516	898	1246	(9)
(10)		CDD10.0	1027	0	0	0	7	81	245	337	271	83	3	0	0	(10)
(11)		CDD18.3	186	0	0	0	0	5	44	87	46	4	0	0	0	(11)
(12)		CDH23.3	1838	0	0	0	3	96	518	733	422	66	1	0	0	(12)
(13)		CDH26.7	562	0	0	0	0	25	172	232	119	13	0	0	0	(13)
(14)	Wind	WSAvg	3.2	2.5	2.8	3.4	4.1	4.0	3.2	3.0	2.9	3.4	3.4	3.0	2.4	(14)
(15)	Precipitation	PrecAvg	346	4	3	5	13	24	57	100	82	35	11	6	6	(15)
(16)		PrecMax	620	19	12	23	48	64	148	264	193	107	44	18	20	(16)
(17)		PrecMin	119	0	0	0	0	0	3	2	19	1	0	0	0	(17)
(18)		PrecStd	94	3	3	5	11	18	33	49	40	24	8	5	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.2	-3.1	9.1	22.6	29.7	34.1	33.8	32.5	28.9	19.2	7.5	-6.1	(19)
(20)			MCWB	-9.2	-4.6	2.8	9.6	13.7	17.5	19.1	18.9	15.1	9.5	2.0	-7.2	(20)
(21)		2%	DB	-13.0	-6.3	5.1	18.3	26.0	30.7	31.3	29.7	25.3	15.3	3.1	-9.7	(21)
(22)			MCWB	-13.7	-7.5	0.3	7.8	12.4	16.7	19.4	17.9	14.4	7.4	-0.4	-10.6	(22)
(23)		5%	DB	-15.5	-8.7	2.4	15.0	23.0	28.5	29.3	27.7	22.5	12.5	0.3	-12.1	(23)
(24)			MCWB	-16.0	-9.8	-0.9	5.7	11.3	16.2	18.8	17.6	13.3	5.9	-2.2	-12.9	(24)
(25)		10%	DB	-17.6	-11.0	0.1	11.9	20.5	26.3	27.2	25.6	19.8	9.9	-2.3	-14.3	(25)
(26)			MCWB	-18.1	-11.8	-2.4	4.4	10.3	15.7	18.2	17.1	12.2	4.4	-4.0	-14.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.2	-4.5	3.2	10.2	14.8	20.0	22.4	21.1	18.1	10.0	2.3	-7.3	(27)
(28)			MCDWB	-8.2	-3.1	8.4	21.4	25.9	27.0	29.4	27.7	24.8	18.0	7.1	-6.2	(28)
(29)		2%	WB	-13.6	-7.5	0.9	8.1	13.4	18.6	21.0	20.0	15.7	8.0	-0.3	-10.6	(29)
(30)			MCDWB	-13.1	-6.3	4.4	17.7	23.4	26.4	27.3	26.3	21.1	14.1	2.9	-9.7	(30)
(31)		5%	WB	-16.0	-9.8	-0.8	6.2	12.1	17.6	20.1	19.0	14.4	6.4	-2.1	-12.8	(31)
(32)			MCDWB	-15.5	-8.8	2.1	14.0	21.3	25.5	26.4	24.6	20.7	11.7	0.1	-12.2	(32)
(33)		10%	WB	-18.1	-11.8	-2.4	4.6	10.9	16.7	19.3	18.1	12.9	4.9	-4.1	-14.9	(33)
(34)			MCDWB	-17.7	-11.0	0.1	11.4	19.4	24.2	25.1	23.4	19.3	9.5	-2.3	-14.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	10.2	10.6	11.1	12.4	11.7	10.2	10.9	11.8	10.6	9.2	8.1	(35)
(36)			MCDBR	10.1	11.5	11.6	16.2	16.8	15.4	13.6	14.0	15.7	14.5	11.6	9.2	(36)
(37)			MCWBR	9.6	10.5	8.3	8.4	7.5	5.5	4.6	5.2	7.0	8.2	8.6	8.6	(37)
(38)		5% WB	MCDWB	10.0	11.6	11.2	15.2	14.6	12.4	11.0	10.7	12.7	13.4	11.2	9.2	(38)
(39)			MCWBR	9.6	10.6	8.2	8.4	7.0	5.2	4.8	4.8	6.5	8.0	8.6	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.279	0.327	0.407	0.475	0.461	0.427	0.396	0.346	0.319	0.287	0.265	(40)	
(41)		taud	2.193	2.144	2.096	2.038	1.910	2.075	2.242	2.295	2.369	2.342	2.259	2.170	(41)	
(42)		Ebn at Noon	795	862	873	833	788	805	829	840	851	810	747	719	(42)	
(43)		Edh at Noon	76	103	130	155	184	157	131	119	99	85	71	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.76	3.02	4.55	5.42	6.08	6.30	5.84	5.34	4.24	2.90	1.77	1.29	(44)	
(45)		RadStd	0.12	0.15	0.22	0.33	0.37	0.46	0.44	0.41	0.28	0.16	0.10	0.15	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	-0.85	-2.87	-3.94	N/A	N/A	N/A	+307	+334	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	-2.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page